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Association of Deforestation With Severe Fever With Thrombocytopenia Syndrome

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Background. Severe fever with thrombocytopenia syndrome (SFTS) virus was first isolated in China in 2009 and has since spread to several Asian countries. SFTS is closely related to environmental factors that accelerate vector growth. We evaluated the associations of SFTS and deforestation with environmental variables.

Methods. For this observational study, we generated multiple Poisson models using national SFTS outbreak data (2013–2018) and official environmental data for Korea. We included established risk factors as variables. Deforestation was used as the main variable. All variables were analyzed according to their spatial characteristics using the R-INLA package.

Results. SFTS cases increased over time and peaked in 2017, at 272, followed by a decrease in 2018. Disease mapping showed a high incidence of SFTS nationwide, with particular risks in Gangwon and Gyeonggi Provinces in the north, and Jeju in the south of South Korea. Deforestation was significantly associated with a higher risk of SFTS in the final model (relative risk, 1.751 [95% confidence interval, 1.125–2.743]).

Conclusions. SFTS outbreaks are associated with deforestation. Therefore, deforestation in Gyeonggi, Gangwon, and Jeju provinces of South Korea needs to be considered in vector-control strategies and active surveillance of SFTS occurrence.

Keywords. SFTS; deforestation; One Health; severe fever with thrombocytopenia syndrome; ticks.

Severe fever with thrombocytopenia syndrome virus (SFTSV) was first isolated in China in 2009 [1] and has since spread to several Asian countries including Japan (2012) [2], Korea (2013) [3], Vietnam (2017) [4], Taiwan (2019) [5], Myanmar (2019) [6], and Thailand (2020) [7]. In Korea, 13 confirmed cases occurred in 2013 [3] and the incidence has increased over time. The Korea Disease Control and Prevention Agency (KDCA) reported 866 cases up to 2018 (Figure 1). In Korea, SFTSV is designated as a reportable infectious disease. Various prevention campaigns have been implemented, including ones encouraging the wearing of long-sleeved clothes and washing immediately after arriving indoors. However, the number of cases is increasing annually, and the incidence varies regionally, which warrants close surveillance.

Severe fever with thrombocytopenia syndrome (SFTS) is a serious emerging infectious disease similar to scrub typhus. Both diseases are transmitted by tick bites and have high case fatality rates (SFTS, 5.2%–32.6% [8, 9]; scrub typhus, 6% [untreated] and 1.4% [treated] [10]). They also share symptoms such as fatigue, headache, dizziness, and fever [11–15], making their discrimination difficult [15]. However, SFTS can be transmitted by contact with infected blood [16], unlike scrub typhus. The risk of person-to-person transmission is high for healthcare workers and patients' family members [17]. Although prompt diagnosis and supportive clinical care can be helpful, because there is no vaccine for SFTS or scrub typhus it is important to preemptively identify and respond to risk factors.

Risk factors for SFTS include working as a farmer, participating in agricultural activity, spring to late summer season, older age, cattle ownership, temperature of 23°C–29°C, and high relative humidity [18–21]. In addition, companion animals serve as reservoirs for SFTS transmission [22]. Working as a farmer and agricultural activity are risk factors [20, 23] because such individuals are more likely to spend time in agricultural or woody areas. Late spring to summer (May–July) is the period of greatest risk [23], and the tick population peaks in May and June [24]. Higher relative humidity and temperature are also risk factors, but extremely high temperatures can hinder transmission by affecting the tick ecosystem [19].

Because SFTS is a vector-borne disease, understanding the tick ecosystem is vital. Temperature, topography, land use, season, and environmental conditions affect tick population growth. For instance, the temperature is higher in hilly provinces than other regions [25], making them good habitats for

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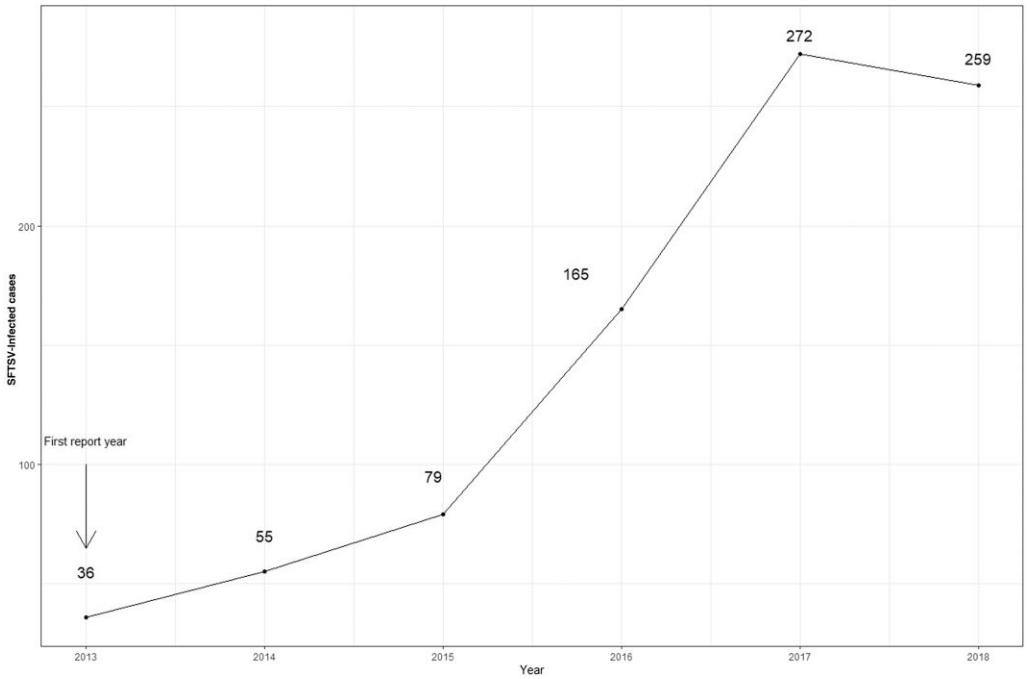

Figure 1. Cases of severe fever with thrombocytopenia syndrome virus (SFTSV) infection from 2013 to 2018.

ticks. The SFTSV vector *Haemaphysalis longicornis* is present in all 8 provinces of South Korea [26]. The optimal temperature for *H longicornis* is 20°C–24°C [27], which typically occurs in grasslands and gravesites during spring (April) to late summer (September) [24]. *Haemaphysalis longicornis* is the dominant vector of SFTS; however, the prevalence of infected ticks is low, 0.2% among those collected from vegetation [28]. In contrast, the seroprevalence of SFTSV is high in domestic animals and wild animals [29–32] (11.1%–28.5%), making them reservoirs of the disease and implying that multiple factors are related to SFTS infection.

Environmental conditions can be risk factors for SFTS because they affect the tick life cycle. Deforestation, defined as forest loss by logging for commercial purposes, affects wildlife and virus circulation [33]. First, as human–wild animal contacts increase, so will the incidence of human infection. The likelihood of such contact is increased by deforestation, as wildlife search for new habitats. Second, increased deforestation reduces biodiversity by changing the ecosystem of the reservoir and vector, and contact with the vector transmits disease [34]. Ticks are commonly detected in small mammals, which serve as secondary vectors or temporary reservoirs [35]. However, as ecosystems are altered by human intervention, virus circulation is affected. Ticks seek substitute reservoirs to replace wildlife including, potentially, animals such as cattle or poultry. The risk of human infection is increased by the practice of animal husbandry. Globally, forest cover is inversely related to outbreaks of vector-borne diseases [34], and increased deforestation is also associated with scrub typhus infection [36]. Therefore, surveillance of SFTS requires an integrated approach considering environmental variables (Figure 2).

We assessed the association of SFTS with deforestation based on established risk factors. We conducted a spatiotemporal analysis of district-level national data on the number of SFTS cases and degree of deforestation in Korea.

METHODS

We examined socioeconomic, geographic, meteorological, and ecological variables (Supplementary Table 1). Using regional data from the KDCA, this observational study analyzed spatiotemporal trends of SFTSV cases from 2013 to 2018, and the association with deforestation.

Variables

Outcome Variable: SFTS Incidence

We used SFTS cases as the outcome variable. The KDCA SFTS database [37] provides aggregated information on confirmed and suspected cases by sex, age, and district. A confirmed case was defined as SFTSV isolation in the presence of symptoms, or a >4-fold increase in antibody titer in recovery compared to acute-phase serum. A suspected case was defined as symptoms with a suspected epidemiological association [38]. We used the 2018 data as the baseline when calculating the incidence rate. We applied regional population data to compare actual and expected cases by year and district. The number of

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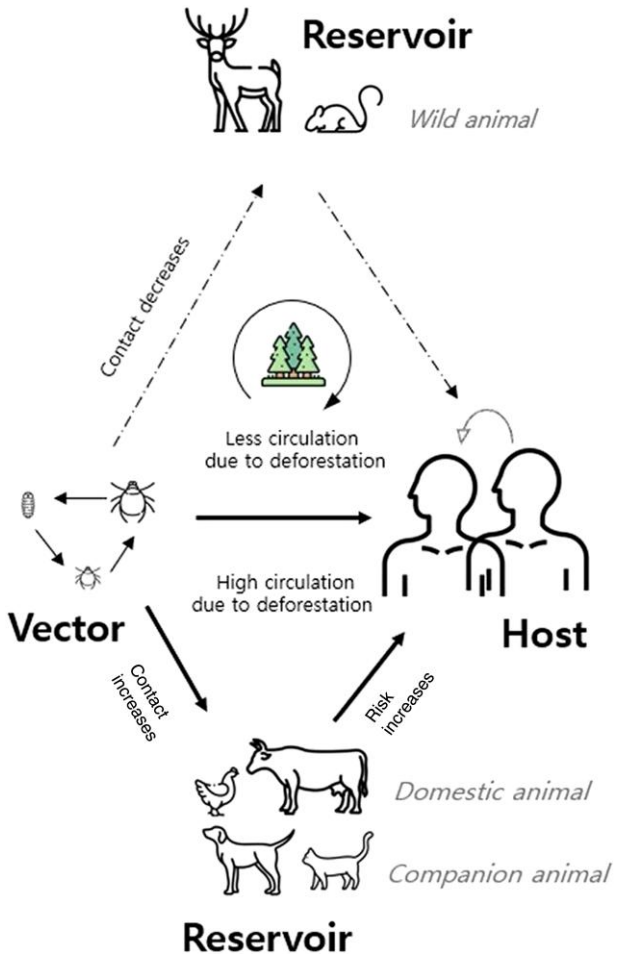

Figure 2. Transmission route of severe fever with thrombocytopenia syndrome virus.

expected cases was estimated by multiplying the standardized incidence rate by the population of each district and year, and was used as the offset variable in the integrated nested Laplace approximation (INLA) formula.

Main Explanatory Variable: Deforestation

Deforestation was the main explanatory variable. Annual tree loss data were obtained from the website of Hansen et al [39] and extracted from satellite images [40]. Using a classification algorithm, Hansen et al [39] captured the forested area in year 2000 and estimated annual deforestation based on that estimate. The raster data of tree loss are composed of approximately 30 × 30 m pixels. Tree loss in each pixel is presented as a binary variable, defined in comparison with the level of tree cover in 2000, and summed for each administrative district. Deforestation variables were acquired for all years, and deforestation levels for 250 districts were categorized into quartiles and scored as 0–3 in each year. The scores for each district were summed from 2013 to 2018, with the final cumulative deforestation levels being 0–18. For statistical analysis, cumulative deforestation levels were categorized into quartiles and subjected to regression analysis using 0–25% deforestation as the reference.

Covariates and Analysis

We used Korean Statistical Information Service [41] data for the total population, number of farmers, and total district area to assess the association between SFTS incidence and deforestation. Meteorological variables such as the annual mean maximum temperature, annual relative humidity and total precipitation were obtained from the Automatic Synoptic Observation System of the Korea Meteorological Administration [42].

Using the ordinary kriging method, we extracted interpolated values for each district of South Korea [43]. Several administrative boundaries were unified and subsequently reseparated during the study years. For example, Cheongju-si originally consisted of Sangdang-gu and Heungdeok-gu in 2013. In 2014, however, when Cheongju-gun was merged into Cheongju-si, 2 new districts were generated: Cheongwon-gu and Seowon-gu. Cheongju-si now comprises 4 districts: Sangdang-gu, Heungdeok-gu, Cheongwon-gu, and Seowon-gu. We modified the data to reflect these changes by using the 2014 district populations and reanalyzed the population and area data for 2013 in the same manner.

For the analysis, we generated descriptive statistics for the main variables and covariates. The SFTS incidence is presented as annual totals. Covariate data are presented as mean values. With the kriged and aggregated dataset, we checked the correlations of the variables by constructing a correlation matrix (Supplementary Figure 1) and calculating the variance inflation factor, and excluded highly correlated variables such as agricultural area and forest coverage from the statistical analysis, as described previously. Bayesian regression models with INLA were used to explore the associations of environmental variables with SFTS. INLA estimates the covariates' effect considering the spatiotemporal association in the model, by including the space-time parameter [44]. To analyze count data, we used a Poisson regression model with expected case as the offset after applying a logarithm. Additionally, due to concerns about and zero cases, we considered negative binomial (NB), zero-inflated Poisson (ZIP), and zero-inflated negative binomial (ZINB) regression models. We used the Watanabe–Akaike information criterion (WAIC) [45] to select the final model. For the analysis, we used the R-INLA R package (version 4.1). The regression results are presented as relative risks (RRs) with 95% confidence intervals (CIs).

RESULTS

Disease mapping showed that SFTS was distributed nationwide, especially in Gangwon and Gyeonggi provinces in the north and Jeju Province in the south of South Korea

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(Figure 3). Deforestation was severe throughout South Korea, and Gangwon, Gyeonggi, and Jeju provinces had high estimates of deforestation and SFTS (Figure 4 and Supplementary Table 2). We created a scatter plot of the association between SFTS incidence and deforestation, in which overlapping observations are represented by circle sizes. The plot suggests a linear relationship between the 2 variables (Supplementary Figure 2). Table 1 lists annual changes in general characteristics. SFTS cases were calculated for each year, and an overall total was obtained. SFTS cases increased over time and peaked in 2017 (272 cases), then slightly decreased in 2018. Annual deforestation data are shown in Table 1; deforestation was severe in individual years (Table 1) and overall (Supplementary Figure 3). The population density and number of farmers decreased over time. The mean maximum temperature, precipitation, and relative humidity were consistent throughout the study period. Mean data by district are presented in Supplementary Figure 4.

The associations between environmental variables and SFTS outbreaks were analyzed by the INLA method (Table 2). According to the estimated WAIC, we used NB as the final model, which showed the lowest estimates. Cumulative deforestation from 2013 to 2018 was associated with a significantly increased risk of SFTS in all the models. Because we used the lowest level of deforestation as the reference, all comparative quartile values were higher. Greater deforestation was associated with a higher risk of SFTS. Urban area was negatively associated with SFTS in the NB model (RR, 0.846 [95% CI, .751–.951]) and the other models. Population density was significantly associated with a decreased risk of SFTSV infection in the NB model (RR, 0.542 [95% CI, .377–.773]), and in the other models. Being a farmer significantly increased the risk of SFTS in the NB model (RR, 1.016 [95% CI, 1.003–1.029]) and ZINB model (RR, 1.015 [95% CI, 1.002–1.028]); however, it was not significant in the Poisson and ZIP models. Mean maximum temperature was negatively associated with SFTS in the NB model (RR, 0.881 [95% CI, .779–.997]), ZINB model (RR, 0.878 [95% CI, .776–.993]), and ZIP model (RR, 0.885 [95% CI, .787–.997]). Precipitation was not significantly associated with SFTSV infection. Relative humidity decreased the risk of SFTSV infection, albeit not significantly.

DISCUSSION

We evaluated trends of SFTS infection and its association with environmental variables by spatiotemporal analysis. SFTS cases gradually increased and peaked in 2017, and slightly decreased in 2018. In the INLA model adjusted for population

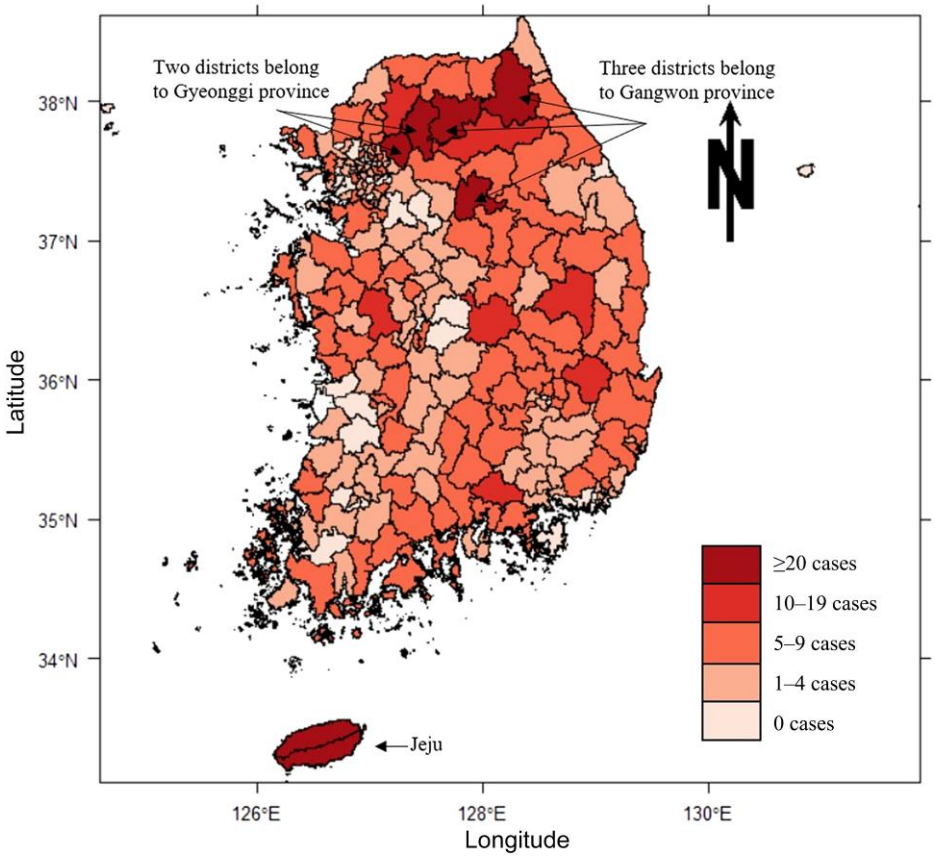

Figure 3. Distribution of total severe fever with thrombocytopenia syndrome cases (mean, 3.46 cases) from 2013 to 2018, by region.

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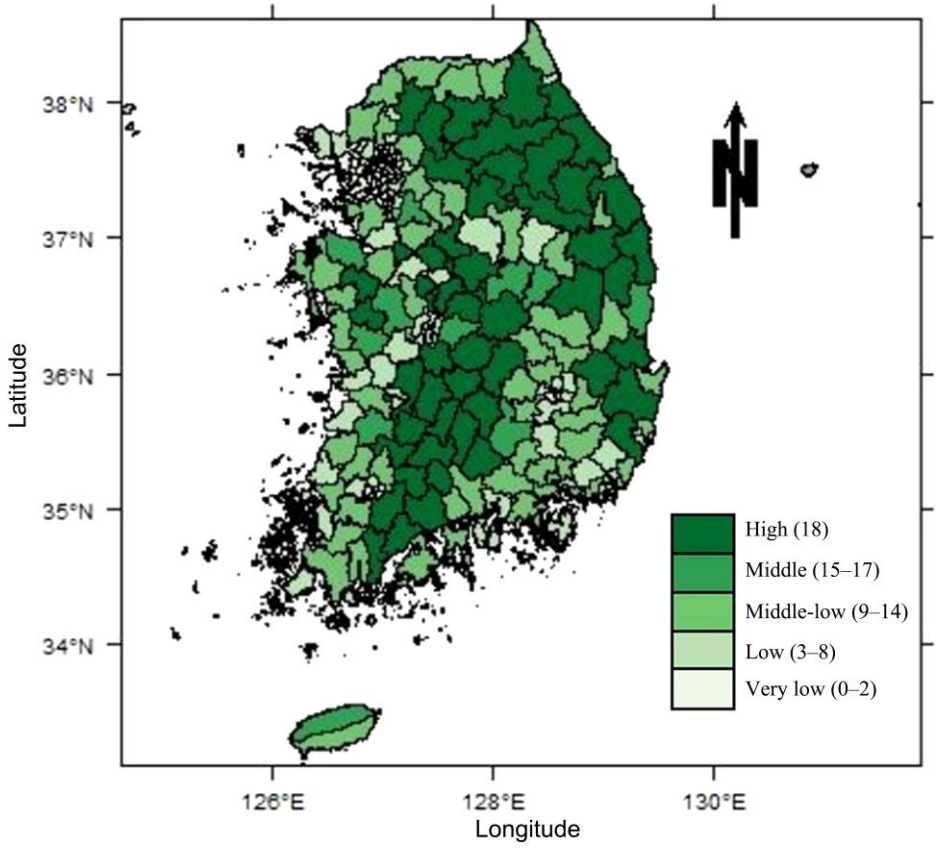

Figure 4. Distribution of cumulative deforestation from 2013 to 2018 by region. Number of cases is shown (mean, 8.99 cases).

Table 1. Descriptive Data for the Variables of Interest by Year

	Year					
Variable	2013	2014	2015	2016	2017	2018
SFTS, No. of cases	36	55	79	165	272	259
Deforestation, km^2	493.208 ± 833.75	713.448 ± 1144.09	936.80 ± 1593.94	1203.076 ± 1869.76	1111.976 ± 1775.38	1158.82 ± 1790.57
Urban areas, 10^8 per km^2	0.683 ± 0.77	0.691 ± 0.78	0.692 ± 0.78	0.692 ± 0.78	0.693 ± 0.78	0.710 ± 0.78
Population density, 10^4 per km^2	0.404 ± 0.61	0.402 ± 0.61	0.400 ± 0.61	0.398 ± 0.60	0.395 ± 0.59	0.392 ± 0.59
Agriculture area, 10^4 per km^2	0.381 ± 0.50	0.369 ± 0.49	0.363 ± 0.47	0.358 ± 0.47	0.346 ± 0.46	0.338 ± 0.45
Farmers, 10^3	1.138 ± 1.03	1.100 ± 1.00	1.027 ± 0.90	0.998 ± 0.87	0.968 ± 0.85	0.926 ± 0.82
Mean maximum temperature, °C	18.109 ± 1.21	18.512 ± 0.76	18.795 ± 0.73	18.894 ± 0.78	18.582 ± 1.06	18.446 ± 0.91
Mean precipitation, 10^3 per mm	1.206 ± 0.16	1.136 ± 0.31	0.934 ± 0.23	1.243 ± 0.30	1.009 ± 0.19	1.402 ± 0.16
Relative humidity, %	68.110 ± 4.14	68.292 ± 2.87	68.540 ± 3.11	68.766 ± 3.50	66.084 ± 3.54	67.520 ± 3.22
Forest, 10^4 per km^2	29.127 ± 39.52	29.127 ± 39.52	29.127 ± 39.52	29.127 ± 39.52	29.127 ± 39.52	29.127 ± 39.52

Data are presented as mean ± standard deviation unless otherwise indicated.

Abbreviation: SFTS, severe fever with thrombocytopenia syndrome.

density, occupation (farmer), and weather, deforestation was associated with a significantly higher risk of SFTS. Deforestation was significantly associated with an increased risk of SFTS in the final and other models, indicating the robustness of the models.

We clarified deforestation as a risk factor for SFTS. Districts with greater deforestation, and those nearby, might have more cases, such as Gyeonggi, Gangwon, and Jeju (Figure 4). Therefore, we suggest that additional management strategies are needed for these districts. Also, since those 3 high-risk

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Table 2. Spatiotemporal Analysis Using Regression Models

Variable	Regression Models: Relative Risk (95% CI)			
	Poisson	ZIP	NB	ZINB
Deforestation (%)				
First quartile	Ref	Ref	Ref	Ref
Second quartile	1.735 (1.166–2.605)	1.811 (1.185–2.787)	1.751 (1.125–2.743)	1.771 (1.135–2.780)
Third quartile	**3.145 (2.000–5.003)**	**3.507 (2.173–5.707)**	**3.296 (1.980–5.538)**	**3.341 (2.002–5.620)**
Fourth quartile	**5.487 (3.497–8.719)**	**5.590 (3.458–9.105)**	**5.604 (3.367–9.418)**	**5.622 (3.369–9.457)**
Urban area (10^8 per km^2)	**0.903 (.825–.987)**	**0.927 (.840–1.022)**	**0.846 (.751–.951)**	**0.849 (.755–.954)**
Population density (10^4 per km^2)	**0.515 (.365–.721)**	**0.508 (.356–.718)**	**0.542 (.377–.773)**	**0.541 (.376–.772)**
Farmers (10^3)	1.007 (.997–1.017)	1.002 (.991–1.013)	**1.016 (1.003–1.029)**	**1.015 (1.002–1.028)**
Mean maximum temperature (°C)	0.914 (.817–1.023)	**0.885 (.787–.997)**	**0.881 (.779–.997)**	**0.878 (.776- .993)**
Precipitation (10^3 per mm)	1.222 (.815–1.821)	1.181 (.774–1.788)	1.189 (.757–1.852)	1.187 (.757–1.844)
Relative humidity (%)	0.994 (.962–1.028)	0.996 (.962–1.031)	0.973 (.940–1.008)	0.975 (.941–1.009)
WAIC	2703.57	2650.08	2631.25	2631.78

Values in bold: *P*-value (<0.05). Abbreviations: CI, confidence interval; NB, negative binomial; Ref, reference; WAIC, Watanabe–Akaike information criterion; ZINB, zero-inflated negative binomial; ZIP, zero-inflated Poisson.

districts are tourist destinations, information on SFTS infection should be delivered to both residents and tourists. For example, SFTS information can be provided to tourists in the form of posters in popular tourist spots or in regional television advertisements.

We analyzed the effects of environmental changes on pathogens and vectors. Ecological factors can affect virus transmission, necessitating evaluation of environmental variables. For example, environmental factors such as temperature changes may accelerate tick growth [46]. In this study, there was a positive association of deforestation with SFTS infection. This association can be explained by deforestation-induced changes in the tick ecosystem. Deforestation can have a long- or short-term effect on the ecosystem of ticks. For example, temperature increases with deforestation [47], promoting tick survival (optimal temperature for tick survival is 20°C–24°C, and the mean maximum temperature in South Korea is 18.1°C–18.9°C). Next, deforestation can cause forests to be substituted for cropland or pasture [48]. This increases human–tick contacts, because host animals and agricultural occupation are risk factors for SFTS infection. Therefore, monitoring of deforestation is vital to control SFTS infection. Also, the long- and short-term effects of deforestation should be tracked.

Several risk factors are related to our findings. Farmers for whom cultivating or harvesting crops, breeding livestock, and preserving or developing agricultural land was significant were included in the final model, but they were not significant in other models. Further studies could focus on specific types of farmers and their occupations to observe transmission routes more precisely. A temperature of 23°C–29°C is reportedly associated with SFTS outbreaks and we observed a negative association between the mean maximum temperature and outbreaks. We presume that this result was not attributable only to the season, and considered the mean maximum temperature instead of mean temperature. SFTSV infection mostly occurred from April to late September, in accordance with prior reports that SFTS incidence peaks in September and October due to a Korean traditional holiday (chu-seok) during which people weed gravesites [25]. Additionally, a higher mean temperature is advantageous for ticks, whereas extreme temperatures hinder their growth [19], suggesting that the effects of summer temperature need to be considered. Population density was negatively associated with SFTS occurrence, possibly because of fewer opportunities for transmission. Also, we presume that districts with low population density are mostly rural and such areas have high proportions of farmers. Precipitation is a risk factor for SFTSV infection [49]; it increased the risk of SFTS in this study, albeit not significantly. Relative humidity was nonsignificantly associated with a decreased infection risk. This may be because the median relative humidity was low (67.18% in 2013–2018), and the optimal relative humidity for tick activity and survival is around 85% [50].

SFTS is an underinvestigated vector-borne disease warranting the One Health approach, which stresses interdependent associations among humans, animals, and the environment. Because tick-borne diseases are associated with complex factors, a multidisciplinary approach involving veterinarians and healthcare professionals is essential. In addition, because the factors are mutually influential, determining the relative contribution of each to SFTS transmission is necessary. For instance, the disease can be transmitted by companion animals [22], and active discussion of its multidimensionally is ongoing; in the future, active surveillance of such animals is required. In South Korea, the One Health approach is being applied to various environmental issues and infectious diseases with cooperation from government ministries and agencies. For example, a national research project is under way to enhance surveillance, amend related legislation, and provide disease-related information to the general public. Furthermore, the effect of deforestation on infectious disease needs to be considered. Deforestation

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has been linked to scrub typhus [36] and SFTS. Therefore, means of slowing deforestation should be included in the comprehensive national development plan, and a strategy for reducing the spread of infectious diseases is needed.

Our findings have several practical implications in terms of reducing SFTS. First, districts with higher deforestation levels and those nearby require improved surveillance to prevent SFTS. Enhanced public awareness of SFTS among not only residents but also tourists is needed, and could be achieved by active promotion and education. Second, a multifactorial approach including pathogen, vectors, and environment is needed to formulate policy on infectious disease. Government agencies should launch a counterplan based on information on vector ecology. Third, we recommend proactive monitoring of areas with severe deforestation. District authorities should verify the preparedness of medical management response teams and appropriate countermeasures.

The strengths of this study include the district-level analysis of SFTS risk taking into consideration multiple variables. To our knowledge, the association between SFTS and deforestation has not been investigated previously. However, this study also had several limitations. First, we included suspected cases of SFTS based on symptoms and epidemiological background, so there may have been some false negatives. Second, the modifiable areal unit problem may have biased the results; however, we resolved this by considering the data of neighboring districts. Third, because we could not evaluate monthly temperature by district, we used the annual mean maximum temperature. Also, because relative humidity has a complicated relationship with temperature and weather patterns, substitute variables such as dew point can be used. Fourth, the SFTS incidence decreased slightly in 2018 (Figure 1), likely because of mixed effects of deforestation; however, long-term investigation using time-series data is needed. Last, although the dominant species of SFTS is *H longicornis,* monitoring of environmental effects on other species is required.

We considered multiple environmental risk factors for SFTSV infection in our spatiotemporal analysis. Deforestation was significantly associated with an elevated risk of SFTS. We suggest that deforestation should be considered in future vector-control strategies, and that active surveillance and proven management strategies that consider deforestation status are needed in Gyeonggi, Gangwon, and Jeju and nearby areas, in Korea.

Supplementary Data

Supplementary materials are available at *The Journal of Infectious Diseases* online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

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Potential conflicts of interest. All authors: No reported conflicts.

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